

Almost-everywhere constancy of the sign on a preconnected target

MathlibAnnex.BilipschitzOrientation.targetJacobianSign_ae_const

theorem

Obtains one constant from the weak equation and preconnectedness of the target.

Statement

Let $n \in \mathbb{N}_0$ and let $E = \mathbb{R}^n$ have the sup norm $\|\cdot\|_\infty$. Let $S, T \subseteq E$ be open, and let $f, g : E \rightarrow E$ be globally Lipschitz maps whose restrictions $f|_S : S \rightarrow T$ and $g|_T : T \rightarrow S$ are mutually inverse. Fix a number $\ell > 0$ such that

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \quad (x, x' \in E).$$

The target T is **preconnected**: it cannot be written as the union of two disjoint, nonempty, relatively open subsets. The empty set is allowed.

Write $Df(x) : E \rightarrow E$ for the Fréchet derivative of f at x when it exists, and set $Df(x) = 0$ at other points. Define the signed Jacobian by $J_f(x) = \det Df(x)$.

Define the source sign $s_f : E \rightarrow \{-1, 1\}$ and its transport $\sigma_f : E \rightarrow \{-1, 1\}$ by

$$s_f(x) = \begin{cases} 1, & J_f(x) \geq 0, \\ -1, & J_f(x) < 0, \end{cases} \quad \sigma_f(y) = s_f(g(y)).$$

In particular, a zero determinant gives sign $+1$, and $|\sigma_f(y)| = 1$ for every y .

Let λ be Lebesgue measure on E . Then there exists $c \in \mathbb{R}$ such that

$$\lambda(\{y \in T : \sigma_f(y) \neq c\}) = 0.$$

Equivalently, $\sigma_f(y) = c$ for Lebesgue-almost every $y \in T$.

Assumptions

The target T is **preconnected**: it cannot be written as the union of two disjoint, nonempty, relatively open subsets. The empty set is allowed. This is an explicit hypothesis in addition to openness of T . No positive-measure or finite-measure hypothesis is imposed.

The sets $S, T \subseteq E$ are open. The inverse and image conditions are

$$\begin{aligned} f(S) &\subseteq T, & g(T) &\subseteq S, \\ g(f(x)) &= x \quad (x \in S), & f(g(y)) &= y \quad (y \in T). \end{aligned}$$

The global bounds use specified constants $C_f, C_g \geq 0$ and $\ell > 0$:

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \leq C_f \|x - x'\|_\infty \quad (x, x' \in E),$$

$$\|g(y) - g(y')\|_\infty \leq C_g \|y - y'\|_\infty \quad (y, y' \in E).$$

The estimates hold on all of E , not just on S or T . The inverse identities are required only on S and T .

Conclusion

A single real number c agrees with the transported Jacobian sign outside a null subset of T . If $T = \emptyset$, every c satisfies this equality; this declaration does not constrain that choice to $\{-1, 1\}$ in the empty case.

The pointwise bound $|\sigma_f| = 1$ supplies local integrability even if T has infinite measure. When T has positive measure, a chosen almost-everywhere constant can be identified with an actual sign value.

Proof route

The sign is locally integrable and satisfies the zero-weak-gradient equation. Apply the constancy theorem with domain T and scalar function σ_f , using the explicit preconnectedness hypothesis.

▼ Proof steps

1. **Supply local integrability and the weak equation.** Since $|\sigma_f| = 1$, every compact set $K \subseteq T$ satisfies

$$\int_K |\sigma_f(y)| d\lambda(y) = \lambda(K) < \infty.$$

The total derivative is measurable, the determinant and its two-valued sign are measurable, and g is continuous. Thus σ_f is measurable and locally integrable on T , as recorded by local integrability of the transported sign. The absolute-value equality is recorded by the source-sign bound and the transported-sign bound.

For every C^1 vector field $W : E \rightarrow E$ that is zero outside a compact subset of T , write $\operatorname{div} W = \operatorname{tr} DW$. Then

$$\int_T \sigma_f(y) \operatorname{div} W(y) d\lambda(y) = 0.$$

This is the zero-weak-gradient theorem for the same maps, target set and Lebesgue measure.

2. **Apply constancy on this target.** The hypotheses now read

$$T \text{ open and preconnected,} \quad \sigma_f \in L^1_{\operatorname{loc}}(T, \lambda), \quad \nabla \sigma_f = 0 \text{ weakly on } T.$$

Apply the preconnected-domain constancy theorem with ambient space $E = \mathbb{R}^n$, Lebesgue measure λ , open set $U = T$, and scalar function $u = \sigma_f$. Openness and preconnectedness are assumptions; Step 1 supplies local integrability and the weak equation. Its output is

$$\exists c \in \mathbb{R} : \quad \lambda(\{y \in T : \sigma_f(y) \neq c\}) = 0,$$

which is exactly the asserted conclusion. This application allows both $n = 0$ and $T = \emptyset$.

Main citations

- Exact declaration and its source —
MathlibAnnex.BilipschitzOrientation.targetJacobianSign_ae_const
- The sign satisfies the weak equation —
MathlibAnnex.BilipschitzOrientation.weakDivergenceZero_targetJacobianSign
- Global constancy on an open preconnected domain —
MathlibAnnex.WeakDivergenceZero.exists_aeConstantOn
- Absolute value of the domain sign —
MathlibAnnex.BilipschitzOrientation.abs_domainJacobianSign
- Absolute value of the transported sign —
MathlibAnnex.BilipschitzOrientation.abs_targetJacobianSign
- Local integrability of the transported sign —
MathlibAnnex.BilipschitzOrientation.locallyIntegrableOn_targetJacobianSign
- Total Jacobian sign, including the zero convention —
MathlibAnnex.BilipschitzOrientation.domainJacobianSign
- Transport of the sign by the specified inverse —
MathlibAnnex.BilipschitzOrientation.targetJacobianSign

Lean source signature (exact)

```
theorem targetJacobianSign_ae_const {n : ℕ}
  (D : BilipschitzOpenData n) (hpre : IsPreconnected D.target) :
  ∃ c : ℝ, AEConstantOn volume (targetJacobianSign D) D.target c
```

In the source	Mathematical meaning
(D : BilipschitzOpenData n)	The record described in this Card: open $S, T \subseteq E = \mathbb{R}^n$, maps $f, g : E \rightarrow E$ restricting to inverse bijections $S \leftrightarrow T$, global upper Lipschitz bounds C_f, C_g , and global lower bound $\ell \ x - x'\ _\infty \leq \ f(x) - f(x')\ _\infty$ with $\ell > 0$. The source name D is not the derivative symbol.
D.source; D.target; D.f; D.g	The local mathematical objects S, T, f, g , respectively, throughout this Card.
targetJacobianSign D	The scalar function $\sigma_f(y) = s_f(g(y))$, where $J_f(x) = \det Df(x)$ and $s_f(x) = 1$ for $J_f(x) \geq 0$, $s_f(x) = -1$ for $J_f(x) < 0$. Zero has sign $+1$ under the cited definitions.
In the source	Mathematical meaning
(hpre : IsPreconnected D.target)	The additional input says that T is preconnected; it permits $T = \emptyset$ and is not stored in the record.
∃ c : ℝ, AEConstantOn volume (targetJacobianSign D) D.target c	The output supplies a real c with $\sigma_f = c$ almost everywhere for $\lambda _T$. This theorem does not return a certificate $c \in \{-1, 1\}$.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 19–25:

```
noncomputable section

open Set MeasureTheory Filter
open scoped BigOperators ENNReal

namespace MathlibAnnex
namespace BilipschitzOrientation
```

▼ Exact content identity

Declaration: `MathlibAnnex.BilipschitzOrientation.targetJacobianSign_ae_const`

Accepted content SHA-256: 66a3ddb7214180414637adb6147bcf1e39e79598327f5cab7379f850f75d28b9

Accepted source guide SHA-256: abf8fd774ca33384b054ab12b0a595ab02009197b7effd1ed1edf9f02540da73

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73